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ORIGINAL ARTICLE

Preimplantation genetic diagnosis for mitochondrial DNA mutations: analysis of one blastomere suffices

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ABSTRACT

Background Preimplantation genetic diagnosis (PGD) is a reproductive strategy for mitochondrial DNA (mtDNA) mutation carriers, strongly reducing their risk of affected offspring. Embryos either without the mutation or with mutation load below the phenotypic threshold are transferred to the uterus. Because of incidental heteroplasmy deviations in single blastomere and the relatively limited data available, we so far preferred relying on two blastomeres rather than one. Considering the negative effect of a two-blastomere biopsy protocol compared with a single-blastomere biopsy protocol on live birth delivery rate, we re-evaluated the error rate in our current dataset.

Methods For the m.3243A>G mutation, sufficient embryos/blastomeres were available for a powerful analysis. The diagnostic error rate, defined as a potential false-negative result, based on a threshold of 15%, was determined in 294 single blastomeres analysed in 73 embryos of 9 female m.3243A>G mutation carriers.

Results Only one out of 294 single blastomeres (0.34%) would have resulted in a false-negative diagnosis. False-positive diagnoses were not detected.

Conclusion Our findings support a single-blastomere biopsy PGD protocol for the m.3243A>G mutation as the diagnostic error rate is very low. As in the early preimplantation embryo no mtDNA replication seems to occur and the mtDNA is divided randomly among the daughter cells, we conclude this result to be independent of the specific mutation and therefore applicable to all mtDNA mutations.

INTRODUCTION

Mitochondrial diseases belong to the most common inborn errors of metabolism.[1] In at least 15% of cases, these multisystem disorders are caused by a primary mutation of the mitochondrial DNA (mtDNA),[1 2] which is exclusively transmitted through the female germline. Female carriers have shown to bear offspring with a wide variety in mutation load, due to the combination of heteroplasmy, meaning a subset of the mtDNAs is mutated, the remainder is wild type and a genetic bottleneck on transmission.[3–7] Given the potential clinical severity of mitochondrial disorders and the general lack of treatment, female carriers may wish to prevent transmission to their (subsequent) offspring. For women wishing to have a genetically related child the two options currently available in clinical practice are prenatal diagnosis (PND)[8–18] and preimplantation genetic diagnosis (PGD).[17 19–25] Mitochondrial replacement techniques are being developed as potential third option[26–29] and will soon be applied to the first human carriers in a licensed and tightly regulated clinical investigation setting (http://www.hfea.gov.uk/9942.html, http://www.hfea.gov.uk/10635.html, accessed on 24 Apr 2017). PND for mtDNA mutations has some drawbacks,[30] the most important one being the potential difficulties in predicting the future child's phenotype based on the mutation load. Therefore, in the majority of cases, PGD is the preferred choice for female carriers. Transmission of the mutation can be prevented by transfer of mutation-free embryos. When these are not available, embryos with mutation loads below a threshold of phenotypic expression are transferred. For some mutations, a specific threshold can be established based on available data,[23] whereas for scarcely reported mutations a generic threshold, calculated by pooling different mtDNA mutations, can be applied.[31] As such, PGD largely reduces the risk of affected offspring rather than fully excluding it. Prerequisites for PGD to be successful in mtDNA carriers are (1) the presence of embryos with mutation load below the threshold of expression and (2) representativeness of the analysed blastomere(s) for the entire embryo. The first has not been an issue so far, but regarding the second point, until now the analysis of two blastomeres has been performed based on incidental heteroplasmy deviations in single blastomeres.[23 24 32] Although blastomere mutation load appears representative for the whole embryo in the majority of cases,[23 24] available data were rather limited. The biopsy of two blastomeres has adverse effects on live birth delivery rate compared with the removal of one cell,[33] and for this reason all nuclear PGD–PCR protocols in our centre have already switched to one-blastomere biopsy for some years now.[34] In the present study, we re-evaluate our two-blastomere biopsy protocol for mtDNA mutations in the context of diagnostic reliability on the current dataset.

METHODS

The error rate was determined in 294 single blastomeres analysed in 73 embryos of 9 female carriers of the m.3243A>G mutation. Single-blastomere mutation loads from the same embryo were compared. Error rate was defined as a false-negative result, that is, a single blastomere with mutation load below the threshold, whereas the other

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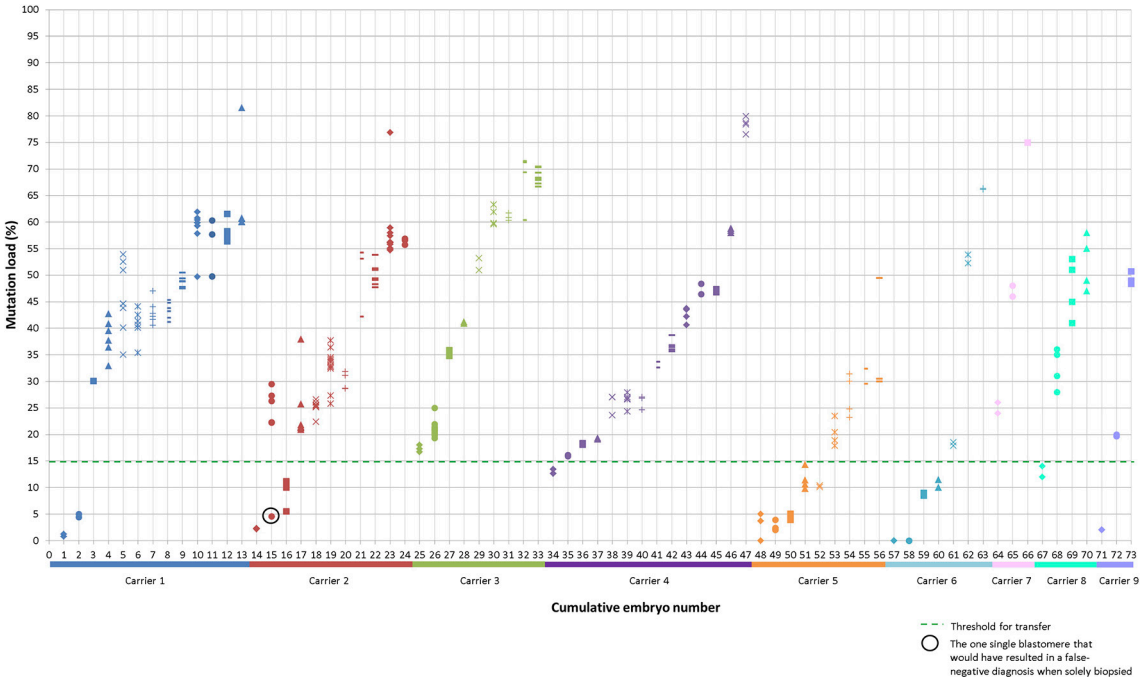


Figure 1 Single blastomere m.3243A>G mutation loads. Each column is one embryo; a total of 73 embryos are plotted. Symbols with the same colour belong to the same carrier. Among these same-coloured symbols, each embryo has a different symbol. Each individual data point represents either a single blastomere, or, in a few cases, multiple single blastomeres when the symbols are so close to each other due to (nearly) identical mutation loads. Only embryos where ≥2 single blastomeres could be biopsied are included in the figure.

one or more blastomeres have heteroplasmy levels above the threshold. The transfer threshold applied in our centre is 15% for the m.3243A>G mutation.[23] The opposite situation, when an embryo is discarded unnecessarily (false-positive result), is less clinically relevant because although a waste and limiting the number of potential pregnancies achieved, this would not result in the birth of an affected child after a PGD treatment. The proportion of single blastomeres potentially resulting in a false-negative misdiagnosis was calculated from the total number of single blastomeres. To perform the analysis, a logistic linear regression model was constructed with a random intercept using a Gaussian mixing distribution to account for the variability between patients or the embryos.

Only embryos of which at least two single blastomeres have been analysed were included. Diagnostic biopsy data (after which embryos with mutation load below the threshold were transferred, meaning that for those embryos only two single blastomere results were available) as well as reanalysis data of embryos unsuitable for transfer were used. Informed consent for reanalysis was obtained from the female carriers. mtDNA mutations other than m.3243A>G were not included in the current analysis because of insufficient sample sizes. A more detailed description of the PGD procedures, the mutation detection/quantification methods and the rationale behind the transfer threshold used, respectively, is provided elsewhere.[23]

RESULTS

In 9 m.3243A>G mutation carriers, a total of 73 embryos underwent 2 or more single-blastomere biopsies (figure 1 and see online supplementary table 1). In figure 1, the concordance between different single blastomeres of the same embryo is depicted for these 73 embryos. Details on for example (1) whether the single blastomere biopsies were diagnostic and/or performed in the context of embryo reanalysis, (2) the destination of the embryos (reanalysis, embryo transfer, frozen) and (3) reproductive outcomes after embryo transfer can be found in online supplementary table 1 . It should be kept in mind that these data are not representative for overall success rates, as for example, embryos where only one blastomere could be biopsied (due to blastomere number <7[23]) are left out here. Among these are embryos with low mutation load which were transferred. This means that pregnancy rates cannot be drawn from these data. That said, our pregnancy rates among mtDNA carriers (data not shown) are in accordance with our large in vitro fertilisation (IVF)/PGD cohorts (personal experience) and do not differ from international numbers.[35] As discussed, pregnancy rates are expected to increase on a single-blastomere biopsy protocol. Regarding the data in online supplementary table 1, it is also important that reanalysis data were only included in the table when they concerned single-blastomere analysis; blastomeres, which could not be dissected in separate blastomeres due to strong intercellular interactions in the morula stage, enabling only the establishment of mean mutation loads of multiple blastomeres, were not included.

For single-blastomere biopsy, an error rate of 0.34% (1/294), with a CI of 0.05%–2.38%, was calculated based on the presence of one blastomere in our series that would have potentially led to a false-negative diagnosis (figure 1). This particular blastomere harboured ~5% mutation load, whereas heteroplasmies of the other blastomeres were between approximately 22% and 30% (figure 1). Notably, if this embryo would have been transferred, its mutation load of <30% would have been associated with a relatively low chance of (severe) symptoms.[31 36] The error rate was zero based on the biopsy of two blastomeres since all potential second blastomeres of this embryo had mutation loads above the threshold and thus would have resulted in a correct diagnosis. The interblastomere heteroplasmy variability between carriers and embryos were both non-significant (with

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respectively p values of 0.954 and 0.642). Although not our primary concern, we did not encounter a false-positive diagnosis in our current dataset.

We apply a transfer threshold of 15%. In some other centres, a threshold of 30% is used instead.[17] Looking at our data, there are six embryos which have blastomeres with mutation loads above and below 30% (embryos 17, 19, 20, 54, 55, 68, figure 1). In the majority of these, it is not clear whether a single blastomere could be qualified as false negative or false positive. Even the biopsy of two blastomeres would mostly not solve this. Another embryo has one blastomere exactly at 30% and two blastomeres above 30%, including a mutation load of almost 50% (embryo 56, figure 1).

DISCUSSION

Our results in m.3243A>G embryos indicated a false-negative error rate of 0.34% for single blastomeres. This error rate is defined as the risk that a biopsied single blastomere does not represent the whole embryo's mutation load and would incorrectly label the embryo as eligible for transfer. Notably, false-positive results were absent in our dataset, although we do not always know the mutation percentages of all blastomeres in the same embryo and/or the fetal and postnatal mutation loads. Our results are generalisable to the population of interest: when comparing the carriers to each other, they did not show significant differences in heteroplasmy variability between blastomeres. The carriers are assumed to reflect the heterogeneity of the population. Furthermore, when comparing all embryos to each other, there were no significant differences in interblastomere variability either, implying that such variability is independent of the carrier. The CI is currently relatively wide (0.05%–2.38%) but that is due to the relatively limited number of data currently available. The low false-negative diagnostic error rate justifies single-blastomere analysis for PGD in m.3243A>G mutation carriers to improve live birth delivery rate. This means we now, based on the expanded dataset compared with our previous study,[23] revise the earlier advice of a 2-blastomere biopsy protocol for mtDNA mutations; the current data show confident enough that one blastomere suffices.

Our conclusions are applicable when using a transfer threshold of 15%. The observation that a threshold of 30% resulted in six embryos with possible diagnostic errors (compared with 1 with the 15% threshold) adds to our arguments to advocate the use of the lower threshold. This takes into account the available data on genotype–phenotype correlations[31 36] and a safety margin for potential (technical) fluctuations in heteroplasmy level.

The fertilised oocyte contains between 50 000 and 550 000 copies of mtDNA.[37] During early embryonic divisions, these copies are divided among daughter cells. In viable embryos, this occurs in a symmetrical way.[38] Meanwhile, the total mtDNA amount seems to remain stable during the first 3 days of preimplantation development (reviewed in reference 34). Therefore, the partitioning of mtDNA among blastomeres in preimplantation embryos is presumed to be a random process. The very high copy number of the fertilised oocyte explains why the Gaussian distribution curves of the single blastomere mutation loads within cleavage-stage embryos (derivable from figure 1) are tight with few deviating values. At the blastomere stage, the mtDNA copies are divided among only six to eight cells, each still containing high amounts. Therefore, if an mtDNA mutation is present, the relative heteroplasmy levels are expected to remain largely the same in each blastomere. This is indeed supported by our data as well as from others.[24] Should certain processes, such as an active copy number reduction (other than by cell divisions), or significant mtDNA replication, occur in the cleavage-stage embryo, this could result in more pronounced interblastomere variability. Both have been reported in other species[39 40] but not in humans. mtDNA replication in early human m.3243A>G preimplantation embryos has been suggested in one study, but this replication could also have occurred in earlier stages (before fertilisation).[41] Moreover, if the observed mtDNA replication indeed took place in the embryo, it may have occurred later in the trophectoderm at the blastocyst stage.[41] This timing corroborates other non-human[42–44] and human[29 37] data. Thus, the available data indicate that no active events influence the mtDNA segregation during the first 3 days of preimplantation development in human embryos, in line with the 'quiet embryo' hypothesis, a characteristic of viable embryos.[45]

Although data are limited, the variation among blastomeres seems in some cases larger than expected by random partitioning only. This could be due to a non-random cytoplasmic distribution of mitochondria and mtDNA within the oocyte or blastomere.[46] Obviously, this is less of an issue in the mutation loads below 15% or above 85% (the latter not being observed for the m.3243A>G mutation). This is indeed illustrated by our data on the m.3243A>G (reference 23 and current report), m.14487T>C (Sallevelt *et al* in preparation) and m.8344A>G[30] mutations. Looking at the m.8993T>G and m.9176T>C mutations, in our experience blastomere mutation load was very constant in three embryos and 18 embryos, respectively, both of a single carrier (reference 23 and unpublished data) with no occasional variability, but numbers are limited. Notably, the majority of these embryos harboured very high mutant loads (~90% and above) and thus rather homogeneous mtDNA populations, for which a more even distribution among daughter cells is more likely on random distribution.

As interblastomere variability seems to be determined by mtDNA partitioning, this is not mutation dependent, implying that our conclusions on the m.3243A>G mutation are generalisable to all mtDNA mutations, known and private mutations alike.

The biopsy of a single blastomere instead of two has positive implications for embryo development and implantation capacity; an overall live birth rate of 37.4% for one-blastomere biopsy versus 22.4% for two-blastomere biopsy was shown.[33] This is the only known biological effect of one-blastomere biopsy versus two-blastomere biopsy. Practically, needing only one blastomere for the mtDNA mutation analysis provides the possibility to analyse two different conditions in two single blastomeres during one PGD treatment. This could be an mtDNA mutation with a nuclear gene defect, which are real examples from our practice (POLG, BRCA2), or an mtDNA mutation with sex selection of male fetuses to definitely stop the transmission to future generations, which has been requested as well. Although technically possible, Dutch legislation currently does not allow selection of embryos based on gender in this situation.

Trophectoderm (TE) biopsy in the blastocyst stage at day 5–6 postfertilisation[47] is expected to combine the possibility of multiple analyses at one hand and more favourable embryo viability with better implantation potential on the other.[48] So far, limited data, both promising[25 49 50] and conflicting,[22 51] are available on TE biopsy for mtDNA mutations and its outcome. The timing of TE biopsy is closer to the 'narrowest' point of the bottleneck, where mtDNA copy number per cell is lowest and consequently the intercell variation highest. Furthermore, evidence from animal studies shows a different mtDNA replication pattern in trophectoderm compared with the inner cell

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mass, from which the embryo will develop.[44] Therefore, considering the biological processes, TE biopsy may be less reliable for mtDNA mutations. As long as no more data are available on TE biopsy for mtDNA mutations and its reliability has not been convincingly shown, it seems preferable to persist with the practice of blastomere biopsy.

Obviously, for homoplasmic mtDNA mutation carriers, PGD cannot select embryos with mutation load below the expression threshold. This may also be the case for carriers with very high mtDNA mutation loads. For those carriers, new techniques (mitochondrial donation) may be an option if ethical and safety requirements are met (http://www.hfea.gov.uk/9942.html). In December 2016, the Human Fertilisation and Embryology Authority (HFEA) has decided to allow these treatments to be carried out in licensed centres, for mtDNA mutation carriers who do not have an alternative such as PGD. Recently, the first clinic (Newcastle Fertility Centre) has been granted this license. This means it can submit patient applications, which the HFEA will consider on a case□by□case basis.

In conclusion, PGD for mtDNA mutations is preferably performed on blastomeres in the cleavage-stage preimplantation embryo. Interblastomere heteroplasmy levels are fairly constant in most preimplantation embryos, but our data show that this specifically applies to the low percentages arguing for a threshold below 15% (m.3243A>G) or 18% (other mtDNA mutations). Here, we specifically show for the m.3243A>G mutation that the biopsy of one blastomere is sufficient for a reliable PGD diagnosis, and we discussed that these results are generalisable for other mtDNA mutations.

Contributors SCEHS: analysed the data of the performed PGD cycles, studied the literature, counseled patients for PGD and wrote the article. JCFMD: coordinated and authorised the genetic laboratory work, documented the PGD data and reviewed the article. EC: involved in the IVF procedures, reviewed the article. ADCP: supervised and authorised the genetic laboratory work, reviewed the article. DMEIH: supervised and authorised the genetic laboratory work, reviewed the article. CEMdD-S: medical coordinator PGD, counselled patients for PGD, reviewed the article. HJMS: head of department Clinical Genomics at the time most analyses were performed, supervised and authorised the genetic laboratory work, supervised writing of the article. PL: performed the statistical analysis, supervised writing of the article. SCEHS, HJMS and PL are responsible for the overall content as guarantors.

Competing interests None declared.

Patient consent Detail has been removed from this case description/these case descriptions to ensure anonymity. The editors and reviewers have seen the detailed information available and are satisfied that the information backs up the case the authors are making.

Ethics approval Medical Research Ethics Committee azM/UM (reference number 03-199)

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